

Risk-Aware Deep Q-Network for Autonomous Navigation in Dynamic Environments

Shaik Mohammad Pardeen
School of CSE & IS,
Presidency University,
Bengaluru-560064, India.
mdpardeen2006@gmail.com

Savanth G
School of CSE & IS,
Presidency University,
Bengaluru-560064, India.
savanthg14@gmail.com

Harika Pudugosula
School of CSE & IS,
Presidency University,
Bengaluru-560064, India.
harika.p@presidencyuniversity.in

Abstract — *Autonomous navigation in dynamic environments remains a significant challenge in reinforcement learning because conventional Deep Q-Network agents typically learn unsafe behaviors only after collisions occur. This paper presents a Risk-Aware Deep Q-Network framework that uses a predictive Gaussian-based spatial risk map for proactive obstacle avoidance. The proposed framework incorporates predicted obstacle trajectories into the agent's state representation and reward function, enabling anticipation of hazardous regions before collisions occur. To improve learning stability and decision-making performance, the framework combines Dueling Deep Q-Network, Double Q-Learning, and Prioritized Experience Replay. Experimental evaluation in a dynamic grid-world environment demonstrates improved reward stability, reduced navigation failures, and higher goal-reaching success rates compared with a standard Deep Q-Network baseline. The results indicate that predictive risk-aware reinforcement learning can enhance proactive navigation behavior and provide a promising foundation for safety-aware autonomous systems operating in dynamic environments.*

Keywords — *Autonomous Navigation, Deep Reinforcement Learning, Deep Q-Network, Risk-Aware Reinforcement Learning, Dynamic Obstacle Avoidance, Dueling Deep Q-Network, Double Q-Learning, Prioritized Experience Replay, Proactive Safety.*

I. INTRODUCTION

Autonomous navigation is a fundamental capability required by intelligent robotic systems operating in dynamic environments. Applications such as autonomous vehicles, warehouse robots, unmanned aerial vehicles, and service robots rely on the ability to perceive their surroundings, plan safe trajectories, and reach target destinations while avoiding obstacles. However, navigation becomes increasingly challenging in environments containing moving objects and uncertain conditions, where obstacle positions continuously change over time. Traditional path-planning and rule-based methods often struggle in such scenarios due to their limited adaptability and inability to anticipate future environmental changes.

Recent developments in Deep Reinforcement Learning (DRL) have significantly advanced autonomous navigation research. Among these approaches, Deep Q-Networks (DQN) have demonstrated the ability to learn navigation

policies directly through interaction with the environment. Several improvements to DQN, including Dueling DQN, Double Q-Learning, and Prioritized Experience Replay (PER), have further enhanced learning stability, convergence speed, and sample efficiency. These techniques have enabled reinforcement learning agents to solve increasingly complex navigation and decision-making problems in dynamic environments.

Mnih et al. introduced the original DQN framework, establishing the foundation for deep reinforcement learning-based navigation. Subsequently, Wang et al. proposed Dueling DQN to improve state-value estimation, van Hasselt et al. developed Double Q-Learning to reduce Q-value overestimation, and Schaul et al. introduced Prioritized Experience Replay to improve learning efficiency. In addition, researchers have explored risk-sensitive reinforcement learning, trajectory prediction, and safe navigation techniques to improve collision avoidance in dynamic environments. Despite these advances, many existing approaches remain reactive, where agents learn unsafe behaviors only after collision events occur.

To address this limitation, this paper proposes a Risk-Aware Deep Q-Network (RA-DQN) framework for proactive autonomous navigation. The proposed method integrates a Gaussian-based spatial risk map generated from predicted obstacle trajectories into both the state representation and reward function of the reinforcement learning agent. Furthermore, the framework combines Dueling DQN, Double Q-Learning, and Prioritized Experience Replay to improve learning stability and navigation performance. By predicting hazardous regions before collisions occur, the proposed approach enables proactive obstacle avoidance and safer navigation behaviour in dynamic environments.

The remainder of this paper is organized as follows. Section II presents the literature review and discusses related work in autonomous navigation and reinforcement learning. Section III describes the proposed Risk-Aware DQN methodology, including environment design, risk-map generation, and reward formulation. Section IV presents the experimental results and performance evaluation. Finally, Section V concludes the paper and outlines potential directions for future research.

II. LITERATURE REVIEW

Autonomous navigation in dynamic environments has traditionally relied on classical path-planning algorithms such as A*, Dijkstra, Artificial Potential Fields, and Velocity Obstacle methods. These approaches are computationally efficient and deterministic but depend heavily on handcrafted rules and predefined environmental assumptions, limiting their effectiveness in crowded and continuously changing environments with moving obstacles [6], [13], [14]. Additionally, traditional navigation systems lack adaptive learning capabilities and struggle to handle uncertainty in complex real-world scenarios.

The emergence of Deep Reinforcement Learning (DRL) significantly advanced autonomous navigation research. Mnih et al. introduced the Deep Q-Network (DQN), which combines deep neural networks with Q-learning to learn action-value functions directly from environment interactions [1]. Despite its success, conventional DQN suffers from unstable convergence, inefficient exploration, and Q-value overestimation. To address these challenges, several enhanced variants were developed. Wang et al. proposed Dueling DQN, which separates state-value and action-advantage estimation to improve learning efficiency [2]. Schaul et al. introduced Prioritized Experience Replay (PER), enabling more informative experiences to be sampled more frequently [3]. Van Hasselt et al. proposed Double Q-Learning, reducing overestimation bias by separating action selection from action evaluation [4]. Together, these improvements enhanced learning stability and sample efficiency.

DRL has been widely applied to autonomous driving, mobile robot navigation, and collision avoidance. Sallab et al. emphasized the importance of reward design, environment representation, and safety constraints in autonomous driving systems [6], while Akmandor et al. developed a DRL-based navigation framework using occupancy-based motion primitives for dynamic obstacle avoidance [12]. Although these approaches achieved promising results, most remain reactive, penalizing agents only after unsafe events or collisions occur. This delayed feedback often causes repeated exploration of dangerous states during training.

To improve safety, researchers have explored risk-sensitive reinforcement learning techniques. Molchanov et al. formulated navigation as a risk-sensitive optimization problem using Conditional Value-at-Risk (CVaR) [5], while Cheng et al. integrated Control Barrier Functions (CBFs) with reinforcement learning to enforce safety constraints [9]. Trajectory prediction methods such as Social LSTM [7] and the Social Force Model [8] further enabled proactive navigation by forecasting future obstacle movements. However, these approaches often require large datasets,

recurrent architectures, or computationally expensive optimization procedures.

Another major challenge is improving robustness and generalization. Tobin et al. introduced domain randomization to enhance sim-to-real transfer [10], while Hausknecht and Stone proposed Deep Recurrent Q-Networks (DRQN) to address partial observability through recurrent memory mechanisms [11]. These studies highlighted the importance of adaptive state representation for real-world deployment.

Despite significant progress, several limitations remain. Many navigation agents still operate reactively, danger is often represented only as binary collision events, and advanced safety frameworks frequently impose high computational costs. To address these challenges, the proposed Risk-Aware Deep Q-Network (RA-DQN) introduces a Gaussian-based spatial risk mapping mechanism integrated directly into the reinforcement learning process. Future obstacle trajectories are predicted using kinematic forecasting, and the resulting spatial risk map is incorporated into both the state representation and reward function. By combining Dueling DQN [2], Double Q-Learning [4], and Prioritized Experience Replay [3], the proposed framework enables proactive collision avoidance while improving learning stability, navigation efficiency, and sample utilization in dynamic environments.

III. PROPOSED METHODOLOGY

The proposed autonomous navigation framework employs a Risk-Aware Deep Q-Network (Risk-Aware DQN) architecture for proactive decision-making and collision avoidance in dynamic environments. Unlike conventional reinforcement learning systems that react only after collisions occur, the proposed framework predicts future obstacle motion and generates a spatial risk representation that guides the agent toward safer navigation trajectories. The methodology integrates dynamic environment modeling, trajectory forecasting, Gaussian-based spatial risk estimation, reinforcement learning optimization, and risk-aware reward shaping into a unified framework.

The system consists of four major components: (1) a dynamic grid-world environment with moving obstacles, (2) a trajectory prediction and spatial risk mapping module, (3) a Risk-Aware Dueling Double Deep Q-Network with Prioritized Experience Replay (PER), and (4) a composite reward mechanism for proactive safety optimization. Future obstacle positions are estimated using kinematic trajectory forecasting, and a Gaussian-weighted spatial risk map is incorporated into both the state representation and reward function to enable predictive collision avoidance. The complete workflow of the proposed Risk-Aware DQN framework is illustrated in Fig. 1

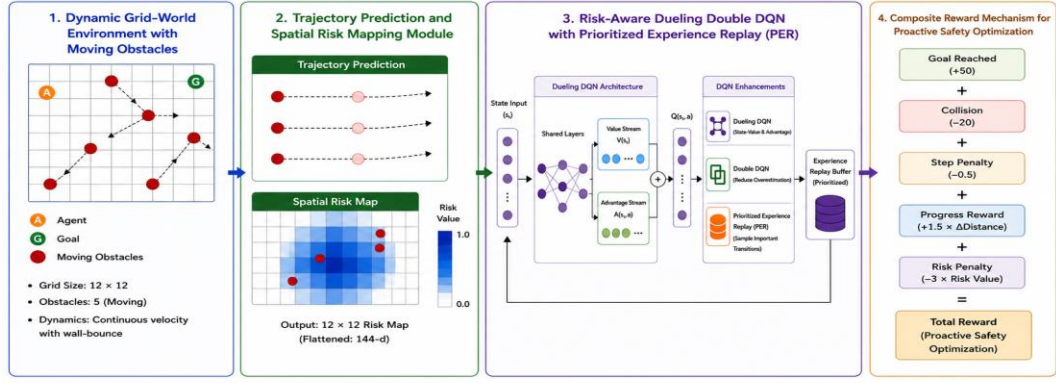


Fig.1. Proposed Risk-Aware DQN Architecture

A. Environment Design

The proposed navigation environment is designed as a dynamic grid-world simulation for autonomous navigation with moving obstacles, as illustrated in Fig. 2. A discrete 12×12 grid structure is used to balance environmental complexity and computational efficiency while maintaining sufficient spatial diversity for obstacle interaction and trajectory planning.

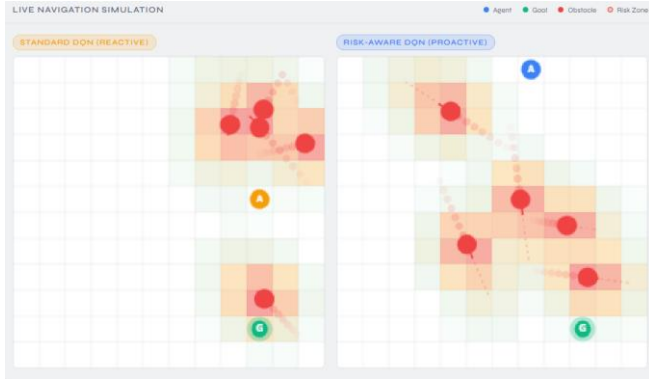


Fig.2. Sample Dynamic Navigation Environment Configurations

The environment contains multiple dynamic obstacles that move continuously using velocity-based motion with wall-bounce dynamics. Obstacle positions are updated at each timestep, creating continuously evolving risk regions that challenge the navigation agent. The agent and goal positions are initialized randomly at the beginning of each episode while maintaining a minimum Manhattan distance to ensure non-trivial navigation tasks.

The agent interacts with the environment using five discrete actions: UP, DOWN, LEFT, RIGHT, and STAY. Each episode terminates when the agent reaches the goal, collides with an obstacle, or exceeds the maximum episode step limit. Historical obstacle trajectories are maintained for future motion estimation and spatial risk map generation. The overall environment configuration used in the proposed framework is summarized in Table I.

TABLE 1. ENVIRONMENT CONFIGURATION

Parameter	Value
Grid Size	12×12
Dynamic Obstacles	5
Action Space	UP, DOWN, LEFT, RIGHT, STAY
Prediction Horizon	5 Steps
Maximum Episode Steps	200
Environment Type	Dynamic Grid World
Reward Mechanism	Risk-Aware Composite Reward

B. State Space Representation

The proposed Risk-Aware DQN framework employs an augmented state representation designed to provide both environmental awareness and predictive safety information to the reinforcement learning agent. The complete state vector consists of 218 dimensions and combines agent information, obstacle states, predicted trajectories, and spatial risk estimation into a unified observation space.

The state representation includes the current agent position and goal position, represented using grid coordinates. Dynamic obstacle information consists of obstacle positions and velocity vectors, enabling the agent to understand both the current and future motion behavior of surrounding obstacles. In addition, predicted obstacle trajectories over a finite prediction horizon are included to provide temporal awareness of future obstacle movement.

A Gaussian-based spatial risk map generated from predicted obstacle trajectories is also incorporated into the state representation. The risk map encodes dangerous regions in the environment as continuous spatial risk values and provides the agent with proactive safety information before collision events occur.

The overall state representation is formulated as a 218-dimensional vector composed of the following components:

- Agent Position: 2 dimensions
- Goal Position: 2 dimensions
- Obstacle Positions: 10 dimensions
- Obstacle Velocities: 10 dimensions
- Predicted Trajectories: 50 dimensions
- Spatial Risk Map: 144 dimensions

The augmented state representation enables the agent to perform proactive navigation and anticipate hazardous regions in dynamic environments.

C. Dynamic Obstacle Modeling

The proposed navigation environment incorporates dynamic obstacles to simulate continuously changing navigation conditions. Each obstacle moves independently using two-dimensional velocity vectors, increasing environmental uncertainty and navigation complexity for the reinforcement learning agent [6]. Obstacle positions are updated at every time step based on their assigned horizontal and vertical velocities.

To maintain continuous motion within the bounded grid environment, wall-bounce dynamics are implemented, where obstacles reverse their motion direction upon reaching environment boundaries [8]. In addition, the framework stores short trajectory histories for each obstacle, which are used by the trajectory prediction module to estimate future obstacle movement [7]. These predicted trajectories are further utilized for Gaussian-based spatial risk map generation and proactive collision avoidance [15].

D. Risk Prediction and Spatial Risk Map Construction

The Risk Prediction Module is the central component of the proposed Risk-Aware DQN framework and is responsible for proactively estimating dangerous regions within the navigation environment. Unlike conventional reinforcement learning systems that penalize the agent only after collisions occur, the proposed framework continuously predicts future obstacle movement and generates a spatial risk representation for proactive navigation [5].

For each dynamic obstacle, future positions are estimated over a finite prediction horizon T using kinematic trajectory forecasting based on the current obstacle position and velocity information [7]. Obstacle motion prediction additionally incorporates wall-bounce dynamics to preserve realistic movement behavior within the bounded grid environment. The future obstacle trajectories are accumulated over multiple time steps to estimate evolving environmental danger.

The future spatial risk associated with obstacle motion is modeled using Gaussian-weighted risk estimation. Each predicted obstacle position contributes a Gaussian-distributed risk value to nearby grid cells, creating a smooth spatial danger representation rather than a binary collision boundary. The Gaussian spatial risk function is defined as:

$$Risk(x, y) = \sum_{obs} w_{obs} \exp\left(-\frac{(x-x_{obs})^2 + (y-y_{obs})^2}{2\sigma^2}\right) \quad (1)$$

Where (x_{obs}, y_{obs}) represents the obstacle position and σ controls the spatial spread of the generated risk distribution.

To model the temporal evolution of environmental danger, the framework accumulates predicted future risks over multiple prediction horizons using temporal decay weighting. Near-future predictions contribute higher risk compared to distant future predictions due to increased prediction reliability. The multi-horizon spatial risk accumulation is computed as:

$$R(x, y) = \sum_{t=0}^T \lambda^t * Gaussian(predicted_{pos_t}, \sigma) \quad (2)$$

Where T denotes the prediction horizon and λ represents the temporal decay factor controlling the contribution of future obstacle predictions.

The accumulated risk values generate a 12×12 spatial risk map representing the dynamic danger distribution across the environment. This risk map is flattened and incorporated into the agent's observation vector, allowing the reinforcement learning agent to recognize potentially unsafe regions before collision events occur. By embedding predictive spatial risk directly into the decision-making process, the proposed framework enables proactive and safety-aware autonomous navigation.

E. Reinforcement Learning Model Used

To improve navigation stability, safety awareness, and learning efficiency, the proposed framework employs a hybrid Risk-Aware Deep Q-Network architecture integrating multiple advanced reinforcement learning techniques. The proposed model is primarily based on the Dueling Deep Q-Network (Dueling DQN) architecture introduced by Wang et al. [2], combined with Double Q-Learning proposed by van Hasselt et al. [4] and Prioritized Experience Replay (PER) developed by Schaul et al. [3]. These techniques collectively improve convergence stability, reduce Q-value overestimation, and enhance sample efficiency in dynamic navigation environments.

Deep Q-Networks (DQN), originally introduced by Mnih et al. [1], demonstrated the ability of deep neural networks to learn action-value functions directly through environmental interaction. However, standard DQN often suffers from unstable convergence and inaccurate Q-value estimation in highly dynamic environments. To overcome these limitations, the proposed framework adopts the Dueling DQN architecture [2], which separates Q-value estimation into two parallel streams: a state-value stream $V(s)$ and an action-advantage stream $A(s, a)$. The final Q-value is computed as follows:

$$Q(s, a) = V(s) + (A(s, a) - \frac{1}{|A|} \sum A(s, a')) \quad (3)$$

This decomposition enables the model to distinguish dangerous environmental states from action-specific effects, improving navigation performance in environments containing moving obstacles and evolving risk conditions.

The network receives a 218-dimensional state vector containing agent position, goal position, obstacle information, predicted obstacle trajectories, and Gaussian spatial risk maps. Shared dense layers first extract high-level navigation features before splitting into value and advantage branches that are later combined to estimate final Q-values for all navigation actions.

To further improve learning reliability, Double Q-Learning [4] is incorporated to reduce Q-value overestimation during target computation. The target Q-value used during training is computed as:

$$Y_t = r_t + \gamma \max_a' Q_{\text{target}}(s_{t+1}, a')(1 - \text{done}) \quad (4)$$

where r_t represents the immediate reward, γ denotes the discount factor, and done indicates terminal states. The temporal-difference (TD) error used for optimization is defined as:

$$\delta = Y_t - Q(s_t, a_t) \quad (5)$$

Additionally, Prioritized Experience Replay (PER) [3] is employed to improve data efficiency by replaying informative transitions more frequently during training. Replay priorities are assigned based on TD-error magnitude as follows:

$$p_i = (|\delta_i| + \epsilon)^\alpha \quad (6)$$

Experiences involving collisions, near-collision events, and successful goal-reaching trajectories therefore receive higher replay priority, enabling the agent to learn more effectively from critical navigation scenarios.

The proposed framework further integrates proactive Gaussian-based spatial risk mapping inspired by risk-sensitive reinforcement learning approaches [5]. The generated risk map is incorporated directly into both the state representation and reward function, enabling the agent to anticipate dangerous regions before collisions occur. The combination of Dueling DQN, Double Q-Learning, PER, and proactive risk-aware navigation results in a lightweight yet effective reinforcement learning framework capable of learning safer and more stable navigation policies in dynamic environments.

F. Reward Function Design

The reward function is designed to encourage safe and efficient autonomous navigation in dynamic environments. The proposed framework employs a composite reward structure combining goal-oriented rewards, collision penalties, progress-based shaping rewards, timeout penalties, and proactive spatial risk penalties. This reward engineering strategy enables the reinforcement learning agent to balance

navigation efficiency with safety-aware decision making [5], [6].

A large positive reward is assigned when the agent successfully reaches the goal position, encouraging long-term task completion similar to reward-driven navigation approaches discussed in deep reinforcement learning literature [1], [6]. In contrast, significant negative penalties are applied for collision events to discourage unsafe navigation behavior in environments containing moving obstacles [5], [15]. Timeout penalties are additionally introduced when the agent fails to reach the goal within the maximum episode duration, preventing inefficient wandering behavior and promoting policy convergence [14].

To improve navigation efficiency, the framework incorporates progress-based reward shaping using the relative change in distance between the agent and the goal position. Positive rewards are assigned when the agent moves closer to the target, while negative rewards are generated when movement increases the distance from the goal. Such reward shaping strategies are widely used in autonomous navigation and reinforcement learning systems to improve policy learning efficiency [6], [12].

The proposed framework further integrates proactive spatial risk penalties derived from the Gaussian-based risk map. The reward is dynamically penalized according to the predicted collision risk associated with the agent's current position, enabling proactive avoidance of hazardous regions before collision events occur [5], [15]. The risk-adjusted reward formulation is defined as follows:

$$R_{\text{adjusted}} = R - w * \text{Risk}_{\text{Exposure}} * \text{Penalty} \quad (7)$$

Training Process where RRR represents the base navigation reward, w denotes the risk weighting coefficient, and $\text{Risk}_{\text{Exposure}}$ corresponds to the spatial collision risk associated with the current agent position.

The complete reward structure used in the proposed framework is summarized in Table II.

TABLE II. REWARD FUNCTION COMPONENTS

Reward Component	Reward Value	Description
Goal Reached	+50	Goal completion
Collision Penalty	-20	Collision punishment
Timeout Penalty	-5	Episode timeout
Progress Reward	± 0.5	Goal-directed movement
Near-Goal Bonus	+2	Target proximity reward
Risk Penalty	Dynamic	High-risk region penalty

G. Training Process

The proposed Risk-Aware DQN framework is trained through continuous interaction with the dynamic navigation environment over multiple episodes [1], [6]. During training, the agent observes the current environment state, selects navigation actions, receives rewards, and updates the Q-network based on accumulated experiences.

An epsilon-greedy exploration strategy is employed to balance exploration and exploitation during learning [1]. Initially, the agent selects random actions with high probability to explore different navigation behaviors. As training progresses, the exploration rate gradually decreases according to exponential epsilon decay, enabling the agent to increasingly exploit learned navigation policies. The exploration decay is defined as follows:

$$\epsilon_t = \epsilon_{\text{end}} + (\epsilon_{\text{start}} - \epsilon_{\text{end}}) * e^{-t/\tau_{\text{decay}}} \quad (8)$$

where ϵ_{start} , ϵ_{end} represent the initial and final exploration rates, respectively.

During interaction with the environment, each transition (s, a, r, s') is stored in a replay buffer for experience replay training [3]. The replay buffer improves sample efficiency by reusing previously collected experiences and reduces correlation between consecutive training samples. Prioritized Experience Replay further improves learning by sampling transitions according to TD-error importance [3].

To stabilize Q-value learning, the framework employs separate online and target networks using Double Q-Learning [4]. The online network is updated continuously during training, while the target network is updated periodically using soft target updates. The target network update is computed as follows:

$$\theta_{\text{target}} \leftarrow \tau \theta_{\text{online}} + (1 - \tau) \theta_{\text{target}} \quad (9)$$

Where τ (tau) denotes the soft update rate controlling parameter synchronization.

Each training episode consists of four major stages:

1. Environment initialization and state observation
2. Action selection using epsilon-greedy policy
3. Reward computation and replay buffer storage
4. Q-network optimization and target network update

Training continues until the agent reaches the goal, collides with an obstacle, or exceeds the maximum episode length [5], [13].

H. Hardware and Software Setup

The proposed Risk-Aware DQN framework was implemented using Python as the primary programming

language, with NumPy utilized for numerical computation, matrix operations, and environment simulation. Visual Studio Code (VS Code) was used as the primary coding and development environment for implementing the reinforcement learning framework. The experiments were executed on a Windows-based system equipped with an Intel Core i5 processor and 8 GB RAM. All reinforcement learning training, Gaussian-based spatial risk-map generation, trajectory prediction, and navigation simulations were performed entirely on CPU without dedicated GPU acceleration. Additional libraries were used for visualization, environment management, and performance evaluation, enabling efficient implementation and reproducible experimentation for the proposed autonomous navigation framework.

IV. EXPERIMENTAL RESULTS

A. Training Performance Analysis

The proposed Risk-Aware DQN framework demonstrated improved navigation stability and learning performance in the dynamic obstacle environment. The training reward curve shown in Fig. 3 indicates that the agent gradually learned safer navigation policies and achieved higher cumulative rewards over training episodes. During the initial stages, the agent exhibited unstable behavior due to random exploration and limited environmental awareness. As training progressed, the framework showed more stable convergence and improved proactive obstacle-avoidance capability compared to the standard DQN baseline.

The training loss curve presented in Fig. 4 illustrates the convergence behavior of the proposed framework during optimization. Initially, higher temporal-difference (TD) errors produced larger loss values due to unstable policy estimation. Over time, the loss gradually decreased as the agent improved its Q-value approximation and navigation strategy. The overall performance demonstrates that integrating Dueling DQN, Double Q-Learning, Prioritized Experience Replay, and Gaussian-based spatial risk modeling contributed to improved learning stability and safer autonomous navigation behavior in dynamic environments.

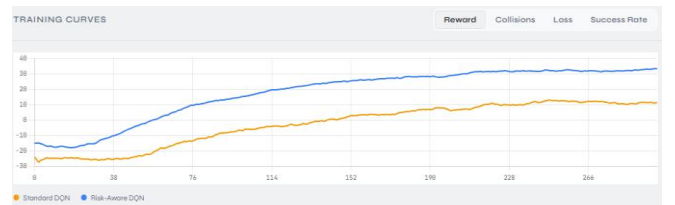


Fig.3. Training reward comparison between Standard DQN and Risk-Aware DQN

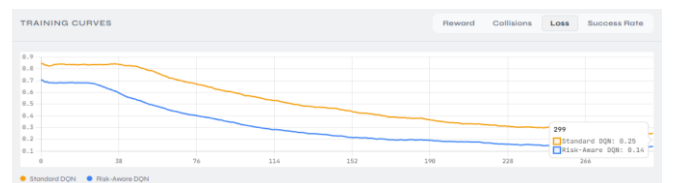


Fig.4. Training Loss comparison between Standard DQN and Risk-Aware DQN

B. Goal Achievement and Collision Analysis

The proposed Risk-Aware DQN framework was evaluated using key navigation performance metrics including success rate, collision rate, and average episode reward. The experimental results demonstrate that integrating proactive Gaussian-based spatial risk modeling significantly improved navigation safety and goal-reaching performance compared to the standard DQN baseline.

The Risk-Aware DQN achieved a higher success rate by learning safer and more efficient navigation policies in the presence of moving obstacles. Additionally, the framework significantly reduced collision frequency due to proactive risk-aware decision making. The average episode reward also improved, indicating enhanced navigation stability and better long-term policy optimization.

The quantitative performance comparison between the Standard DQN and the proposed Risk-Aware DQN framework is summarized in Table III.

TABLE III. Performance comparison between Standard DQN and Risk-Aware DQN over the last 50 training episodes.

Metric	Standard DQN	Risk-Aware DQN
Average Reward	12.7	32.5
Collision Rate	1.21	0.34
Success Rate(in %)	58%	78%

C. Risk-Aware Behavior Analysis

The proposed Risk-Aware DQN framework demonstrated proactive obstacle-avoidance behavior by utilizing Gaussian-based spatial risk prediction during navigation. Unlike conventional reactive DQN systems that respond only after approaching dangerous regions, the proposed framework continuously predicted future obstacle trajectories and avoided high-risk areas before collision events occurred. This proactive risk-aware strategy enabled the agent to maintain safer navigation paths in dynamic environments containing moving obstacles. The generated Gaussian-based spatial risk heat map shown in **Fig. 6** illustrates how dangerous regions surrounding predicted obstacle trajectories were assigned higher spatial risk values.

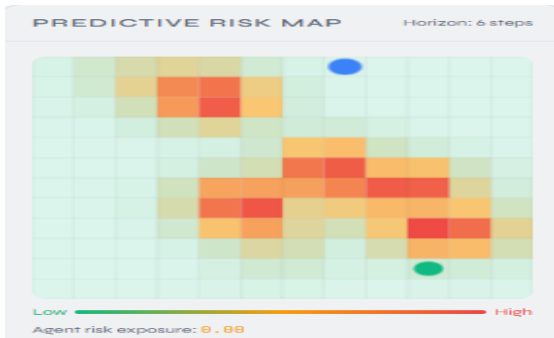


Fig. 6. Predictive Gaussian-Based Spatial Risk Heat map

The generated risk maps allowed the reinforcement learning agent to identify evolving danger distributions across the environment and select safer navigation trajectories. High-risk regions surrounding predicted obstacle trajectories were assigned larger risk values, guiding the agent toward safer and more stable paths. The navigation comparison presented in **Fig. 7** further demonstrates that the proposed Risk-Aware DQN produced smoother and safer trajectories compared to the standard reactive DQN baseline. While the conventional DQN frequently entered dangerous regions due to reactive decision making, the proposed framework maintained greater distance from predicted obstacle motion paths and achieved more stable goal-reaching behavior. These results indicate that proactive spatial risk modeling improved collision avoidance capability, navigation stability, and overall autonomous decision-making performance.

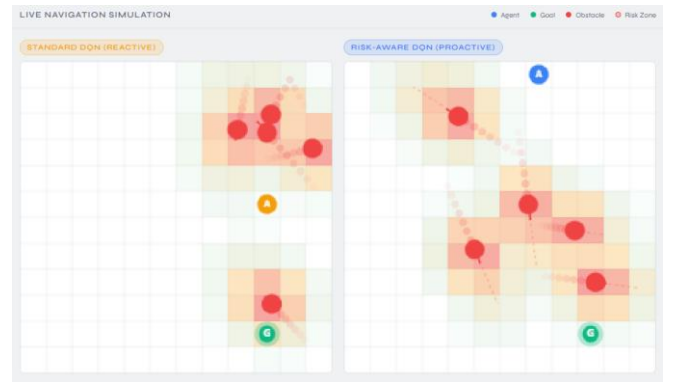


Fig. 7. Navigation Comparison between Standard DQN and Risk-Aware DQN

IV. I. COMPARATIVE RESULTS

The proposed Risk-Aware DQN framework was compared against a conventional Standard DQN baseline to evaluate the impact of proactive spatial risk modeling on autonomous navigation performance. The comparison focused on key reinforcement learning metrics including average reward, collision rate, success rate, best episode reward, and worst episode reward.

Experimental results demonstrate that the proposed Risk-Aware DQN significantly improved navigation safety and policy stability compared to the standard DQN model. The framework achieved higher average rewards and success rates while substantially reducing collision frequency in dynamic obstacle environments. In addition, the proposed model exhibited improved worst-case episode performance, indicating greater robustness and reduced catastrophic navigation failures during training.

The quantitative comparison between the Standard DQN and the proposed Risk-Aware DQN framework is summarized in Table IV.

TABLE IV. COMPARATIVE PERFORMANCE ANALYSIS

Metric	Standard DQN	Risk-Aware DQN
Average Reward	12.7	32.5
Collision Rate	1.21	0.34
Success Rate (in %)	58%	78%
Best Episode Reward	18.4	53.6
Worst Episode Reward	-37.5	-21.5

V. EXPERIMENTAL RESULTS

I. Safety Evaluation

The proposed Risk-Aware DQN framework demonstrated improved safety performance through proactive collision avoidance and spatial risk-aware navigation. Experimental results showed that the framework significantly reduced collision frequency compared to the standard DQN baseline by avoiding predicted high-risk regions before collision events occurred. The lower collision rate indicates that integrating Gaussian-based spatial risk prediction improved the agent's ability to navigate safely in dynamic environments containing moving obstacles.

J. Efficiency Evaluation

The proposed framework achieved improved navigation efficiency by learning shorter and safer trajectories toward the goal location. The agent required fewer unnecessary movements and demonstrated faster convergence during training due to proactive risk-aware decision making. Compared to the conventional DQN model, the Risk-Aware DQN achieved more stable reward progression and improved goal-reaching efficiency in complex dynamic environments.

K. Stability Evaluation

The proposed Risk-Aware DQN framework exhibited improved learning stability throughout training. Reward fluctuations gradually decreased as the agent learned safer navigation policies and improved Q-value estimation. Additionally, the framework reduced catastrophic navigation failures such as repeated collisions and unstable exploration behavior. The integration of Dueling DQN, Double Q-Learning, and Prioritized Experience Replay contributed to more stable reinforcement learning convergence in continuously changing navigation scenarios.

L. Sample Efficiency

The proposed framework demonstrated improved sample efficiency by learning effective navigation behavior using prioritized replay-based training. Prioritized Experience Replay enabled the agent to learn more effectively from

informative experiences such as collision events and successful goal-reaching trajectories. As a result, the Risk-Aware DQN achieved faster policy improvement and more efficient learning compared to the standard reactive DQN baseline.

VI. CONCLUSION

Autonomous navigation in dynamic environments remains a challenging problem due to continuously changing obstacle motion, environmental uncertainty, and the reactive limitations of conventional reinforcement learning systems. Traditional DQN-based navigation approaches often rely on collision-based feedback, resulting in delayed safety awareness and unstable navigation behavior in complex environments. To address these limitations, this work proposed a Risk-Aware Deep Q-Network framework integrating Gaussian-based spatial risk mapping, trajectory prediction, Dueling DQN, Double Q-Learning, and Prioritized Experience Replay to improve navigation safety, learning stability, and decision-making efficiency. By incorporating predicted environmental risk directly into both the state representation and reward function, the proposed framework transformed collision avoidance from a reactive process into a proactive risk-aware navigation strategy.

Experimental evaluation demonstrated that the proposed Risk-Aware DQN achieved improved reward stability, reduced collision frequency, higher success rates, and more stable convergence behavior compared to the standard DQN baseline. The generated spatial risk maps enabled the agent to anticipate dangerous regions before collision events occurred, resulting in safer and smoother navigation trajectories in dynamic obstacle environments. The proposed framework provides a lightweight and effective foundation for safety-aware reinforcement learning in applications such as autonomous vehicles, warehouse robotics, drone navigation, and intelligent robotic systems. Future research can further enhance the framework through PyTorch-based implementation, recurrent trajectory prediction models, safety-constrained reinforcement learning, and real-world autonomous navigation deployment.

REFERENCES

- [1] V. Mnih, K. Kavukcuoglu, D. Silver, A. A. Rusu, J. Veness, M. G. Bellemare, A. Graves, M. Riedmiller, A. K. Fidjeland, G. Ostrovski, and D. Hassabis, "Human-Level Control Through Deep Reinforcement Learning," *Nature*, vol. 518, no. 7540, pp. 529–533, Feb. 2015.
- [2] Z. Wang, T. Schaul, M. Hessel, H. van Hasselt, M. Lanctot, and N. de Freitas, "Dueling Network Architectures for Deep Reinforcement Learning," in *Proc. Int. Conf. Machine Learning (ICML)*, New York, NY, USA, 2016, pp. 1995–2003.
- [3] T. Schaul, J. Quan, I. Antonoglou, and D. Silver, "Prioritized Experience Replay," in *Proc. Int. Conf. Learning Representations (ICLR)*, San Juan, Puerto Rico, 2016.
- [4] H. van Hasselt, A. Guez, and D. Silver, "Deep Reinforcement Learning with Double Q-Learning," in *Proc. AAAI Conf. Artificial Intelligence*, Phoenix, AZ, USA, 2016, pp. 2094–2100.

- [5] A. Molchanov, E. Schmerling, and M. Pavone, "Risk-Sensitive Reinforcement Learning for Autonomous Navigation," *IEEE Robot. Autom. Lett.*, vol. 5, no. 2, pp. 3451–3458, Apr. 2020.
- [6] A. E. Sallab, M. Abdou, E. Perot, and S. Yogamani, "Deep Reinforcement Learning Framework for Autonomous Driving," *Electron. Imaging*, vol. 2017, no. 19, pp. 70–76, Jan. 2017.
- [7] A. Alahi, K. Goel, V. Ramanathan, A. Robicquet, L. Fei-Fei, and S. Savarese, "Social LSTM: Human Trajectory Prediction in Crowded Spaces," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Las Vegas, NV, USA, 2016, pp. 961–971.
- [8] D. Helbing and P. Molnar, "Social Force Model for Pedestrian Dynamics," *Phys. Rev. E*, vol. 51, no. 5, pp. 4282–4286, May 1995.
- [9] R. Cheng, G. Orosz, R. M. Murray, and J. W. Burdick, "End-to-End Safe Reinforcement Learning Through Barrier Functions for Safety-Critical Continuous Control Tasks," in *Proc. AAAI Safe Reinforcement Learning Workshop*, Honolulu, HI, USA, 2019.
- [10] J. Tobin, R. Fong, A. Ray, J. Schneider, W. Zaremba, and P. Abbeel, "Domain Randomization for Transferring Deep Neural Networks from Simulation to the Real World," in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots Syst. (IROS)*, Vancouver, BC, Canada, 2017, pp. 23–30.
- [11] M. Hausknecht and P. Stone, "Deep Recurrent Q-Network," in *Proc. AAAI Fall Symp. Sequential Decision Making Intelligent Agents*, Arlington, VA, USA, 2015.
- [12] N. U. Akmandor, H. Li, G. Lvov, and K. Karydis, "Deep Reinforcement Learning Based Robot Navigation in Dynamic Environments Using Occupancy Values of Motion Primitives," *arXiv preprint arXiv:2208.08034*, 2022.
- [13] Z. Xie, "Towards Safe Navigation Through Crowded Dynamic Environments," in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots Syst. (IROS)*, Prague, Czech Republic, 2021.
- [14] H. Le, S. Saeedvand, and C.-C. Hsu, "A Comprehensive Review of Mobile Robot Navigation Using Deep Reinforcement Learning Algorithms in Crowded Environments," *J. Intell. Robot. Syst.*, vol. 111, no. 2, pp. 1–28, 2024.
- [15] F. Hart, "Enhanced Method for Reinforcement Learning Based Dynamic Obstacle Avoidance by Assessment of Collision Risk," *Neurocomputing*, vol. 568, p. 127097, 2024.
- [16] K. Mokhtari and A. R. Wagner, "Safe Deep Q-Network for Autonomous Vehicles at Unsignalized Intersection," *arXiv preprint arXiv:2106.04561*, 2021.
- [17] H. Taheri, S. R. Hosseini, and M. A. Nekoui, "Deep Reinforcement Learning with Enhanced PPO for Safe Mobile Robot Navigation," *arXiv preprint arXiv:2405.16266*, 2024.
- [18] G. Nehme and T. Y. Deo, "Safe Navigation: Training Autonomous Vehicles Using Deep Reinforcement Learning in CARLA," *arXiv preprint arXiv:2311.10735*, 2023.
- [19] W. Fei, "Deep-Reinforcement-Learning-Based UAV Autonomous Navigation and Collision Avoidance," *Chin. J. Aeronaut.*, vol. 37, no. 1, pp. 112–126, 2024.
- [20] Y. Wang, "Autonomous Navigation Control and Collision Avoidance Using Dueling Deep Q-Networks," *J. Marine Sci. Eng.*, vol. 13, no. 11, p. 2108, 2025.